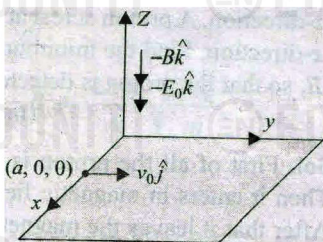


SUBJECTIVE QUESTIONS

1. A positively charged particle of mass m and charge q is projected on a rough horizontal x - y plane surface with z -axis in the vertically upward direction. Both electric and magnetic fields are acting in the region and given by $\vec{E} = -E_0\hat{k}$ and $\vec{B} = -B_0\hat{k}$, respectively. The particle enters into the field at $(a_0, 0, 0)$ with velocity $\vec{v} = v_0\hat{j}$. The particle starts moving in some curved path on the plane. If the coefficient of friction between the particle and the plane is μ , calculate the

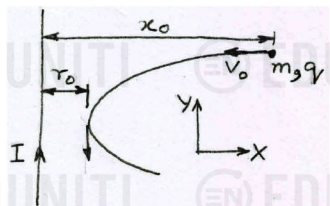
- time when the particle will come to rest
- distance travelled by the particle when it comes to rest



2. A positively charged particle having charge $q_1 = 1 \text{ C}$ and mass $m_1 = 40 \text{ gm}$ is revolving along a circle of radius $r_1 = 40 \text{ cm}$ with velocity $v_1 = 5 \text{ m s}^{-1}$ in a uniform magnetic field with center of circle at origin O of a three-dimensional system. At $t = 0$, the particle was at $(0, 0.4\text{m}, 0)$ and velocity was directed along positive x direction. Another particle having charge $q_2 = 1 \text{ C}$ and mass $m_2 = 10 \text{ g}$ moving uniformly parallel to positive z -direction with velocity $v_2 = 40/\pi \text{ m s}^{-1}$ collides with revolving particle at $t = 0$ and gets stuck to it. Neglecting gravitational force and coulomb force, calculate x -, y - and z - coordinates of the combined particle at $t = (\pi/40)\text{s}$.

3. An electron of charge ' $-e$ ' and mass m accelerated by a potential difference V first enters into a uniform electric field of a parallel plate capacitor whose plates extend over a length l in the direction of initial velocity. The electric field is normal to the direction of initial velocity and its strength varies with time as $E = \alpha t$, where α is a constant. Then the electron enters into a uniform magnetic field of induction B . Direction of magnetic field is same as that of the electric field. Calculate pitch of helical path traced by the electron in the magnetic field. (Mass of electron, $m = 9 \times 10^{-31} \text{ kg}$)

4.



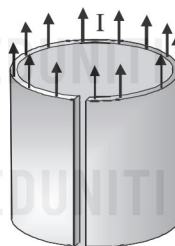
A long straight wire carries a current I . A particle having a positive charge q and mass m , kept at distance x_0 from the wire is projected towards it with speed v_0 . Find the closest distance of

approach r_0 of charged particle to the wire.

5. A current I flows along a round loop. Find the integral $\int \vec{B} \cdot d\vec{r}$ along the axis of the loop within the range from $-\infty$ to $+\infty$. Explain the result obtained

6. A square loop of wire, edge length a , carries a current I . Compute the magnitude of the magnetic field produced at a point on the axis of the loop at a distance x from the centre.

7.

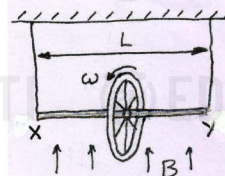


A current I flows along a thin walled tube of radius R with a long longitudinal slit of width b ($b \ll R$). What is the magnitude of magnetic field induction at a point inside the tube at a distance r from the slit?

8. Three infinitely long wires, each carrying current I in the same direction, are in the x - y plane of a gravity free space. The central wire is along the y -axis while the other two are along $x = \pm d$.

- If the central wire is displaced along the z -direction by a small amount and released, show that it will execute simple harmonic motion. If the linear mass density of the wire is λ , find the time period of oscillation.

9.



A ring of radius R having uniformly distributed charge Q is mounted at the centre of a rod XY of length L and suspended by two identical strings. The tension in strings in equilibrium is T_0 . Now a vertical magnetic field is switched on and ring is rotated at constant angular velocity ω . Find the maximum ω with which the ring can be rotated if the strings can withstand a maximum tension of $3T_0/2$.

10. A sphere of radius R , uniformly charged with the surface charge density σ , rotates around the axis passing through its center at an angular velocity ω .

- Find the magnetic field at the center of the rotating sphere.
- Find its magnetic moment.

MORE THAN ONE CORRECT QUESTIONS

11. A positively charged particle with velocity $\vec{v} = x\hat{i} + y\hat{j}$ moves in a magnetic field $\vec{B} = y\hat{i} + x\hat{j}$. The force acting on the particle has magnitude F . Which of the following statement(s) is/are correct?
- (a) If $x = y$, no force will act on charged particle
- (b) If $x > y$, $F \propto (x^2 - y^2)$
- (c) If $x > y$, force will act along z -axis
- (d) If $y > x$, force will act along x -axis

12. A particle of specific charge $q/m = \pi \times 10^{10} \text{ C kg}^{-1}$ is projected from the origin along the positive x -axis with a velocity of 10^5 ms^{-1} in a uniform magnetic field $\vec{B} = -2 \times 10^{-3} \hat{k}$ tesla. Which of the following statements is/are correct?
- (a) The centre of the circle lies on the y -axis
- (b) The time period of revolution is 10^{-7} s .
- (c) The radius of the circular path is $5/\pi \text{ mm}$
- (d) The velocity of the particle at $t = 0.25 \times 10^{-7} \text{ s}$ is $10^5 \hat{j} \text{ m/s}$

13. Velocity and acceleration vector of a charged particle moving in a uniform magnetic field at some instant are $\vec{v} = 3\hat{i} + 4\hat{j}$ and $\vec{a} = 2\hat{i} + x\hat{j}$. Select the correct options
- (a) $x = -1.5$ (b) $x = 3$
- (c) Magnetic field is along z -direction
- (d) Kinetic energy of the particle is constant

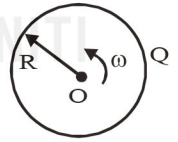
14. A charged particle P leaves the origin with speed $v = v_0$ at some inclination with the x -axis. There is a uniform magnetic field B along the x -axis. P strikes a fixed target T on the x -axis for a minimum value of $B = B_0$. P will also strike T if
- (a) $B = 2B_0$, $v = 2v_0$ (b) $B = 2B_0$, $v = v_0$
- (c) $B = B_0$, $v = 2v_0$ (d) $B = B_0/2$, $v = 2v_0$

15. From a cylinder of radius R , a cylinder of radius $R/2$ is removed as shown in the figure. Current flowing in the remaining cylinder is I . Magnetic field strength is



- (a) zero at point A
- (b) zero at point B
- (c) $\frac{\mu_0 I}{3\pi R}$ at point A
- (d) $\frac{\mu_0 I}{3\pi R}$ at point B

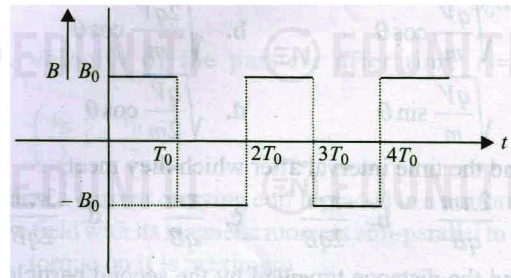
16. A nonconducting disc having uniform positive charge Q , is rotating about its axis with uniform angular velocity ω . The magnetic field at the centre of the disc is
- (a) directed outward



- (b) having magnitude $\frac{\mu_0 Q \omega}{4\pi R}$
- (c) directed inwards
- (d) having magnitude $\frac{\mu_0 Q \omega}{2\pi R}$

PARAGRAPH TYPE QUESTIONS**Paragraph for Question Nos. 17 to 18**

In a region, the magnetic field along x -axis changes according to the graph given in fig.



The time period, radius and pitch of the helical path are T_0 , R_0 and $3R_0$ respectively. The particle is projected at an angle θ_0 with the positive x -axis toward positive y -axis in x - y plane.

17. The coordinates of the particle at $t = 3T_0/2$ are
- (a) $(9R_0/2, 0, -2R_0)$ (b) $(3R_0, 0, -R_0)$
- (c) $(9R_0/2, 0, 2R_0)$ (d) $(3R_0, 0, R_0)$
18. What is the maximum possible distance between any two positions during a time interval of one time period, i.e., during $t_0 \leq t \leq t_0 + T_0$?
- (a) $3R_0$ (b) $4R_0$
- (c) $5R_0$ (d) $6R_0$

Paragraph for Question Nos. 19 to 21

In a certain region of space, there exists a uniform and constant electric field of magnitude E along the negative y -axis of a coordinate system. A charged particle of mass m and charge q ($q > 0$) is projected from the origin with speed $2v$ at an angle of 60° with the positive x -axis in x - y plane. When the x -coordinate of particle becomes $\sqrt{3}mv^2/qE$, a uniform and constant magnetic field of strength B is also switched on along positive y -axis. Let $t = 0$ at this instant.

19. Velocity of the particle just before the magnetic field is switched on is

- (a) $v\hat{i}$ (b) $v\hat{i} + \frac{\sqrt{3}v}{2}\hat{j}$
- (c) $v\hat{i} - \frac{\sqrt{3}v}{2}\hat{j}$ (d) $2v\hat{i} - \frac{\sqrt{3}v}{2}\hat{j}$

20. x -coordinate of the particle as a function of time after the magnetic field is switched on is

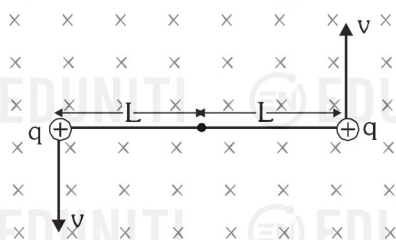
(a) $\frac{\sqrt{3}mv^2}{qE} - \frac{mv}{qB} \sin\left(\frac{qB}{m}t\right)$ (b) $\frac{\sqrt{3}mv^2}{qE} + \frac{mv}{qB} \sin\left(\frac{qB}{m}t\right)$
 (c) $\frac{\sqrt{3}mv^2}{qE} - \frac{mv}{qB} \cos\left(\frac{qB}{m}t\right)$ (d) $\frac{\sqrt{3}mv^2}{qE} + \frac{mv}{qB} \cos\left(\frac{qB}{m}t\right)$

21. z -coordinate of the particle as a function of time after the magnetic field is switched on is

(a) $\frac{mv}{qB} \left[1 - \cos\left(\frac{qB}{m}t\right)\right]$ (b) $-\frac{mv}{qB} \left[1 - \sin\left(\frac{qB}{m}t\right)\right]$
 (c) $-\frac{mv}{qB} \left[1 - \cos\left(\frac{qB}{m}t\right)\right]$ (d) $\frac{mv}{qB} \left[1 - \sin\left(\frac{qB}{m}t\right)\right]$

Paragraph for Question Nos. 22 to 24

Two charged particles each of mass ' m ', carrying charge $+q$ and connected with each other by a massless inextensible string of length $2L$ are describing circular path in the plane of paper. The speed of each particle is $v = qB_0 L / m$ (where B_0 is constant) about their centre of mass. A uniform magnetic field $\vec{B}$ exists into the plane of paper as shown in figure. Neglect any effect of electrical and gravitational forces.



22. The magnitude of the magnetic field such that no tension is developed in the string will be

(a) $B_0/2$ (b) B_0
 (c) $2B_0$ (d) 0

23. If the actual magnitude of magnetic field is half of that calculated above, then tension in the string will be

(a) $\frac{3q^2 B_0^2 L}{4m}$ (b) zero
 (c) $\frac{q^2 B_0^2 L}{2m}$ (d) $\frac{2q^2 B_0^2 L}{m}$

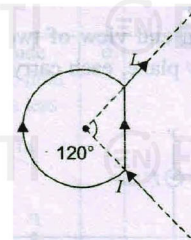
24. Given that the string breaks when the tension is $T = \frac{3q^2 B_0^2 L}{4m}$.

Now if the magnetic field is reduced to such a value that the string just breaks, then find the maximum separation between the two particles during their motion

(a) $16L$ (b) $4L$
 (c) $14L$ (d) $2L$

Paragraph for Question Nos. 25 to 26

In fig., the circular and the straight parts of the wire are made of same material but have different diameters. The magnetic field at the centre is zero.



25. Ratio of the currents I_1 and I_2 flowing through the circular and straight parts is

(a) $\frac{\sqrt{3}}{2\pi}$ (b) $\frac{2\sqrt{3}}{\pi}$
 (c) $\frac{3\sqrt{3}}{2\pi}$ (d) $\frac{3\sqrt{3}}{2\sqrt{2}\pi}$

26. The ratio of the diameters of wires of circular and straight parts is

(a) $\frac{1}{\sqrt{2}}$ (b) $\frac{2\sqrt{3}}{\pi}$
 (c) $\frac{3\sqrt{3}}{2\pi}$ (d) $\sqrt{2}$

Paragraph for Question Nos. 27 to 28

The current density in a linear conductor of radius a varies with r according to the relation $J = kr^2$, where k is a constant and r is the distance of a point from the axis of the conductor.

27. Find the magnetic field induction at a point at distance r from the axis when $r < a$. Assume relative permeability of the conductor to be unity.

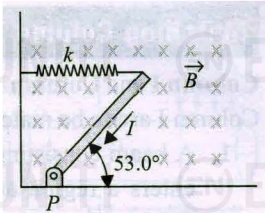
(a) $\frac{\mu_0 k \pi a^4}{4r}$ (b) $\frac{\mu_0 k r^3}{2}$
 (c) $\frac{\mu_0 k \pi a^4}{2r}$ (d) $\frac{\mu_0 k r^3}{4}$

28. Find the magnetic field induction at a point at distance r from the axis when $r > a$. Assume relative permeability of the medium surrounding the conductor to be unity.

(a) $\frac{\mu_0 k a^4}{4r}$ (b) $\frac{\mu_0 k r^3}{2}$
 (c) $\frac{\mu_0 k \pi a^4}{2r}$ (d) $\frac{\mu_0 k r^4}{4}$

Paragraph for Question Nos. 29 to 30

As shown in figure, a thin uniform rod of negligible mass and length 0.2 m is attached to the floor by a frictionless hinge at point P . A horizontal spring with force constant $k = 5 \text{ N m}^{-1}$ connects the other end of the rod to a vertical wall. The rod is in a uniform magnetic field $B = 2 \text{ T}$ directed into the plane of the figure.

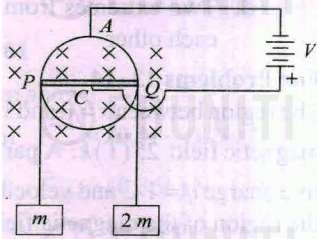


There is current $I = 4 \text{ A}$ in the rod, in the direction shown.

29. The torque about P , due to the magnetic force on the rod is
(a) 0.16 Nm (b) 0.32 Nm
(c) 0.64 Nm (d) 0.28 Nm
30. When the rod is in equilibrium and makes an angle of 53° with the floor, the stretch in the spring is
(a) 1.6 m (b) 0.8 m
(c) 0.4 m (d) 0.2 m

Paragraph for Question Nos. 31 to 33

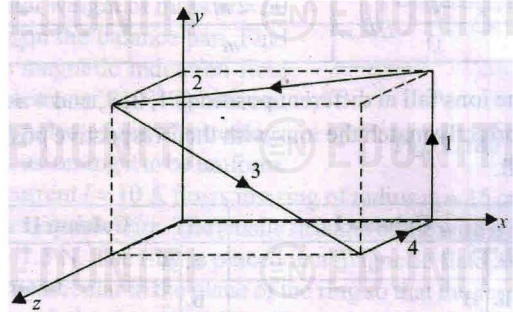
A conducting ring of mass m and radius r has a weightless conducting rod PQ of length $2r$ and resistance $2R$ attached to it along its diameter. It is pivoted at its center C with its plane vertical, and two blocks of mass m and $2m$ are suspended by means of a light inextensible string passing over it as shown in fig. The ring is free to rotate about C and the system is placed in a magnetic field B (into the plane of the ring). A circuit is now completed by connecting the ring at A and C to a battery of emf V . It is found that for certain value of V , the system remains static.



31. In static condition, the current through rod PC is
(a) V/R (b) $V/2R$
(c) $4V/R$ (d) $2V/R$
32. The value of V can be related with m , B and r as
(a) $2mgR/Br$ (b) mgR/Br
(c) $mgR/2Br$ (d) $3mgR/Br$
33. The net torque applied by the tension in strings is
(a) $\frac{3BVR^2}{R}$ (b) $\frac{BVR^2}{R}$
(c) $\frac{BVR^2}{3R}$ (d) $\frac{BVR^2}{2R}$

Paragraph for Question Nos. 34 to 36

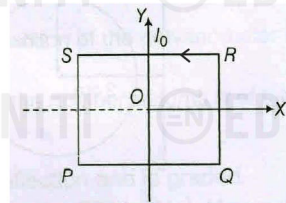
A wire carrying a 10 A current is bent to pass through various sides of a cube of side 10 cm as shown in fig. A magnetic field $\vec{B} = (2\hat{i} - 3\hat{j} + \hat{k}) \text{ T}$ is present in the region.



34. The net force on the loop is
(a) $\vec{F}_{net} = 0$ (b) $\vec{F}_{net} = (0.1\hat{i} - 0.2\hat{k}) \text{ N}$
(c) $\vec{F}_{net} = (0.3\hat{i} + 0.4\hat{k}) \text{ N}$ (d) $\vec{F}_{net} = (0.36\hat{k}) \text{ N}$
35. The magnetic moment vector of the loop is
(a) $(0.1\hat{i} + 0.05\hat{j} - 0.05\hat{k}) \text{ Am}^2$
(b) $(0.1\hat{i} + 0.05\hat{j} + 0.05\hat{k}) \text{ Am}^2$
(c) $(0.1\hat{i} - 0.05\hat{j} + 0.05\hat{k}) \text{ Am}^2$
(d) $(0.1\hat{i} - 0.05\hat{j} - 0.05\hat{k}) \text{ Am}^2$
36. The net torque on the loop is
(a) $-0.1\hat{i} + 0.4\hat{k} \text{ Nm}$ (b) $-0.1\hat{i} - 0.4\hat{k} \text{ Nm}$
(c) $0.2\hat{i} - 0.4\hat{k} \text{ Nm}$ (d) $0.1\hat{i} - 0.4\hat{k} \text{ Nm}$

Paragraph for Question Nos. 37 to 38

A uniform, constant magnetic field B_0 is directed at an angle of 45° to x -axis in xy -plane. $PQRS$ is a rigid square wire frame carrying a steady current I_0 with its centre at origin O . At time $t = 0$, the frame is at rest in the position shown in the figure with its side parallel to x and y -axis. Each wire of the frame has mass M and length L .



37. The torque acting on the frame due to magnetic field and axis about which it will rotate is
(a) $\frac{I_0 L^2 B_0}{\sqrt{2}}(\hat{i} + \hat{j})$ and SQ (b) $\frac{I_0 L^2 B_0}{\sqrt{2}}(-\hat{i} + \hat{j})$ and SQ
(c) $\frac{I_0 L^2 B_0}{2}(-\hat{i} - \hat{j})$ and PR (d) $\frac{I_0 L^2 B_0}{2}(\hat{i} + \hat{j})$ and PR
38. Find the angle by which the frame rotates under the action of this torque in a short interval of time Δt . (Δt is so short that any variation in the torque during this interval may be neglected)



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(a) $\frac{3I_0 B_0}{4M} (\Delta t)^2$

(b) $\frac{4I_0 B_0}{M} (\Delta t)^2$

(c) $\frac{2I_0 B_0}{3M} (\Delta t)^2$

(d) $\frac{4I_0 B_0}{3M} (\Delta t)^2$